

Notes: Bursztyn, Fujiwara & Pallais (AER, 2017)

“Acting Wife”: Marriage Market Incentives and Labor Market Investments

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1 Big picture

- **Research question.** Do single women avoid career-enhancing actions because those actions signal traits that may be penalized in the dating or marriage market, such as ambition, assertiveness, competitiveness, or willingness to take demanding jobs?
- **Core idea.** The same action can be valuable in the labor market and costly in the marriage market. Asking for a raise, volunteering for leadership, speaking up in class, or reporting willingness to work long hours can help professionally, but may also signal a type that some potential partners dislike.
- **Economic mechanism.** A single woman maximizes a joint objective that includes both labor-market returns and marriage-market returns. If an action is observable to potential partners and signals a marriage-market-costly trait, she may strategically choose a lower-career-investment action even when her private labor-market preference is high.
- **Key analogy.** The paper is conceptually similar to social-image models such as “acting white.” Here the social audience is the dating or marriage market, and the distorted behavior is labor-market investment rather than school effort.
- **Setting.** The setting is an elite U.S. MBA program. This is useful because students are making early-career investments, interacting heavily with peers, and many are single or searching for a partner.
- **Four empirical components.** The paper combines: administrative grades, an anonymous survey about avoided workplace behaviors, a primary field experiment with randomized observability of career preferences, and a supplementary field experiment that changes whether single women reveal job choices in front of male or female peers.
- **Observational grade evidence.** Unmarried women perform similarly to married women on less-visible components of grades, such as exams and problem sets, but receive much lower class participation grades. Men do not show the same relationship-status pattern.
- **Survey evidence.** Single women report avoiding behaviors that could help their careers because they do not want to appear too ambitious, assertive, or pushy. The most frequent avoided behavior is taking initiative in negotiating a raise or promotion.
- **Primary experimental evidence.** When single women expect classmates to observe a real-stakes career questionnaire, they report lower desired compensation, less willingness to travel, lower willingness to work long hours, lower tendency to lead, and lower professional ambition. Men and non-single women are mostly unaffected.
- **Supplementary experimental evidence.** Single women report less career-focused job choices when male peers, rather than only female peers, observe their choices. This points directly to the marriage-market audience.
- **Placebo logic.** Writing ability is labor-market-relevant but not expected to be marriage-market-costly. Single women do not reduce their reported writing ability in public, which rules against a broad “public humility” interpretation.
- **Main contribution.** The paper identifies a social-image and marriage-market channel that can distort labor-market investment even among high-ability, highly career-oriented women.
- **Big takeaway.** Gender gaps can arise not only from preferences, skills, discrimination, or family constraints, but also from the private cost of being seen to want career success.

2 Data and measurement

2.1 Overall empirical architecture

The paper uses four complementary datasets from the same elite MBA environment. Each dataset targets a different piece of the mechanism.

- **Administrative grades** provide naturally occurring behavior in a required course.
- **Anonymous survey data** measure whether students self-report avoiding career-helping behaviors for image reasons.
- **Primary field experiment** creates random variation in whether career preferences are expected to be visible to classmates.
- **Supplementary field experiment** creates random variation in whether the observing peers are male or female.

Detailed interpretation: The combination matters. The grades show real behavior in a normal school setting, the survey elicits the mechanism directly, and the experiments isolate observability and audience composition.

2.2 Administrative grades sample

- The grades sample covers the 2010–2016 entering cohorts of the MBA program.
- The course is a required introductory economics class, so sample selection into the course is minimal.
- Grade components include visible class participation and less-visible nonparticipation components such as exams and problem sets.
- The key comparison is married versus unmarried students in administrative data.
- Participation is observed by classmates and can signal assertiveness, confidence, or willingness to speak publicly.
- Exams and problem sets are not similarly visible to classmates, so they are less likely to affect the marriage market.

Detailed interpretation: The grade data provide a natural placebo design. If unmarried women were simply lower ability or less career-oriented, they should also perform worse on exams and problem sets. The fact that the gap appears in visible participation but not in private grade components supports the signaling channel.

2.3 Survey sample and avoided behaviors

- The anonymous survey was administered to first-year MBA students in fall 2016.
- Students were asked whether, in the previous two years, they avoided actions that could help professionally because they might appear too ambitious, assertive, or pushy.
- The listed behaviors were: taking initiative in negotiating a raise or promotion; asking for a leadership role in a team or task force; offering to make a presentation or sales pitch; and speaking up at meetings.
- Relationship status is classified as single versus non-single, where non-single includes serious relationship, cohabiting, engaged, or married.
- The survey is anonymous and therefore cannot be linked to admissions records, but anonymity helps elicit sensitive answers.

Detailed interpretation: The survey is not causal evidence, but it is important because it asks the mechanism directly. It tells us whether the hypothesized trade-off is something students recognize in their own behavior.

2.4 Primary field experiment: observability of career preferences

- The primary experiment occurred on the first day of the MBA program during a career advising session.
- Students completed a career questionnaire that they believed would be used by the career center for internship placement.
- The questionnaire asked about preferences over desired compensation, willingness to travel, desired hours, tendency to lead, professional ambition, comfort in competitive environments, and writing ability.
- The treatment randomized the expected peer audience. In the public condition, students were told that *your* answers would be discussed in class. In the private condition, they were told that *anonymized* answers would be discussed.
- In both treatments, the career counselor would see the answers. Therefore, the professional audience is held constant.
- The treatment changes social visibility to classmates, not whether the career office observes the responses.

Detailed interpretation: This is the central identification device. If single women privately want different jobs, that should show up in both treatments. If they strategically self-present because of marriage-market concerns, the change should occur only when classmates can observe the answers.

2.5 Primary experiment outcomes

The outcome variables are designed to separate marriage-market-costly career signals from neutral or placebo traits.

- **Desired compensation:** a direct proxy for career ambition and willingness to pursue high-paying jobs.
- **Days willing to travel:** captures willingness to accept demanding jobs, especially consulting-type roles.
- **Weekly hours willing to work:** captures willingness to pursue high-intensity occupations.
- **Tendency to lead:** captures leadership/assertiveness.
- **Professional ambition:** captures direct self-presentation of ambition.
- **Comfort in competitive environments:** intended to capture competitiveness, though the paper later argues respondents may not have interpreted it as a negative marriage-market signal.
- **Writing ability:** placebo outcome. It is useful professionally but is not predicted to be penalized in the marriage market.

Detailed interpretation: The outcome design is stronger than a generic career questionnaire because it includes both theoretically targeted variables and placebo variables.

2.6 Supplementary field experiment: male versus female peer audience

- The supplementary experiment was run three months after the primary experiment during a career class.
- Students were placed into groups and asked to choose between pairs of hypothetical jobs.
- The paper focuses on single women and compares all-female peer groups with groups that include male peers.

- The job pairs trade off salary versus hours, promotion prospects versus travel, and social impact versus coworker interaction.
- The first two choices are expected to reveal career ambition or willingness to accept demanding career paths. The third is a placebo-like choice not expected to reveal the same marriage-market-costly career ambition.

Detailed interpretation: The supplementary experiment sharpens the audience interpretation. If observability by classmates matters because classmates include potential male partners, single women should respond more when male peers observe them.

2.7 Relationship status categories

- In the grade data, the comparison is married versus unmarried because administrative records classify marital status that way.
- In the survey and experiments, the comparison is single versus non-single. Non-single includes serious relationship, cohabiting, engaged, or married.
- This distinction matters because a serious relationship may reduce marriage-market signaling incentives even if the person is not legally married.

Detailed interpretation: The paper’s mechanism is about being active in the marriage market, not legal marital status alone. That is why the experiment uses single/non-single rather than only married/unmarried.

3 Models and empirical specifications

3.1 Conceptual model: private labor-market payoff and public marriage-market signal

A useful way to summarize the mechanism is to write a student’s payoff from action a as

$$U(a; v) = L(a) - \mathbf{1}_{\{public\}} M(a, v),$$

where $L(a)$ is the labor-market payoff, $M(a, v)$ is the marriage-market cost of what action a signals to potential partners, and v denotes relationship status or partner-market relevance.

- If the action is private, the student chooses the labor-market maximizing action.
- If the action is public, a single woman may choose a lower career-investment action to reduce the marriage-market signal.
- If the student is not active in the marriage market, the public/private distinction should matter less.
- If the public audience is female-only rather than male peers, the marriage-market cost should be smaller.

Detailed interpretation: The paper does not need a fully estimated structural model. The predictions are sharp: public observability should affect single women on marriage-market-relevant signals, not other groups and not placebo signals.

3.2 Grade comparison

The administrative grade comparison can be summarized as

$$Grade_{ics} = \alpha_s + \beta_1 Unmarried_i + \gamma' X_i + \varepsilon_{ics},$$

where i indexes students, c grade component, and s section or cohort controls.

The key comparison is across grade components:

$$\beta_{Women}^{Participation} < 0, \quad \beta_{Women}^{Nonparticipation} \approx 0,$$

while for men

$$\beta_{Men}^{Participation} \approx 0, \quad \beta_{Men}^{Nonparticipation} \approx 0.$$

Detailed interpretation: The grade specification is not a randomized design, but the contrast across visible and private grade components creates a built-in falsification test. The mechanism predicts a gap only in the visible component.

3.3 Survey comparison

For each avoided behavior b , the paper compares group means:

$$Avoid_{ib} = \alpha + \beta_1 SingleWoman_i + \beta_2 NonSingleWoman_i + \beta_3 SingleMan_i + \varepsilon_i.$$

The central patterns are:

- single women report the highest rate of avoidance;
- non-single women are closer to men than to single women;
- the strongest behavior is negotiation avoidance;
- the survey index of any avoided behavior is also highest for single women.

Detailed interpretation: This survey is motivational evidence. It tells us that the strategic avoidance mechanism is not simply imposed by the researcher; respondents themselves describe avoiding career-helping actions for exactly the hypothesized image reason.

3.4 Primary experiment: treatment effect within each group

The basic treatment-effect equation is

$$Y_i = \alpha + \tau Public_i + \varepsilon_i,$$

estimated separately by gender and relationship status.

A richer pooled version is

$$Y_i = \alpha + \tau Public_i + \delta SingleWoman_i + \theta(Public_i \times SingleWoman_i) + \Gamma' G_i + u_i,$$

where G_i includes gender and relationship-status indicators.

- τ captures the public-treatment effect for the omitted group.
- θ captures whether single women respond differently to public observability.
- A negative θ for career-ambition outcomes supports the marriage-market signaling mechanism.

Detailed interpretation: Random assignment makes public observability causal. The interaction with single-woman status identifies whether the treatment effect is concentrated in the theoretically relevant group.

3.5 Kling-Liebman-Katz index

The paper uses an index that combines outcomes for which high values are labor-market favorable and potentially marriage-market costly:

$$KLK_i = \frac{1}{K} \sum_{k=1}^K \frac{Y_{ik} - \bar{Y}_k}{s_k}.$$

- Outcomes are standardized to place them on comparable scales.
- The index reduces multiple-hypothesis concerns.
- A large negative treatment effect on the index for single women means the overall bundle of public career signals is reduced.

Detailed interpretation: The index is important because no single outcome is definitive. The hypothesis predicts a pattern across several labor-market signals.

3.6 Primary experiment placebo logic

The placebo equation is the same as the main treatment equation, but the outcome is writing ability:

$$Writing_i = \alpha + \tau Public_i + \varepsilon_i.$$

Detailed interpretation: If single women are simply humble in public or more reluctant to self-promote, writing ability should fall too. It does not. This supports the narrower interpretation that only marriage-market-costly career traits are downplayed.

3.7 Supplementary experiment: male-peer audience

The supplementary experiment can be summarized as

$$Choice_{ig} = \alpha_s + \tau MalePeers_g + \varepsilon_{ig},$$

where g indexes the peer group and α_s are section fixed effects.

- The treatment is whether a single woman is in a group with male peers rather than only female peers.
- The first two outcomes are career-focused choices: salary over lower hours, and promotion over less travel.
- The third outcome is a placebo-style choice: social impact over coworker interaction.

Detailed interpretation: This is the most direct test of the marriage-market audience. If male peers are potential partners, single women should shift away from career-intensive choices when those peers observe the choice.

4 Identification and empirical/experimental design

4.1 Core identification problem

The main challenge is distinguishing true private preferences from strategic public self-presentation. A single woman may report lower ambition for many reasons:

- she may truly prefer less travel, fewer hours, or less leadership;

- she may be less confident or more humble;
- she may be responding to norms about femininity;
- she may be concerned about what potential partners infer from her answer.

Detailed interpretation: The paper’s design isolates the last channel by changing observability while holding the underlying labor-market consequences constant.

4.2 Primary experiment identification

The strongest identification comes from the primary experiment.

- Random assignment determines whether answers are expected to be visible to classmates.
- The career counselor sees answers in both treatments, so career stakes are held constant.
- The public/private language differs by one key word: “your” versus “anonymized.”
- Students do not know that there are two versions of the questionnaire.
- Students believe the questionnaire could affect internship placement, so responses have real stakes.

Detailed interpretation: The design changes the social audience without changing the professional audience. Therefore, treatment effects are naturally interpreted as social-image responses rather than changes in actual career incentives.

4.3 Why relationship status is central

The model predicts that single women should be the group most affected by observability.

- Single women are most likely to be concerned about the marriage-market signal sent to peers.
- Non-single women should face weaker dating-market incentives, especially if they are in serious relationships.
- Men should not face the same penalty from displaying ambition because ambition is typically valued rather than penalized in male partners.

Detailed interpretation: Relationship-status heterogeneity is therefore not just a control. It is part of the identification strategy.

4.4 Why the male-peer supplementary experiment matters

The supplementary experiment addresses a possible weakness of the primary experiment: classmates are a broad peer audience, not all of whom are potential partners.

- If the mechanism is general peer pressure, female-only groups could still create pressure.
- If the mechanism is marriage-market signaling, the presence of male peers should matter more.
- The experiment therefore compares all-female groups to groups with male peers.

Detailed interpretation: This is the key audience-composition test. It helps distinguish marriage-market signaling from generic public-image concerns.

4.5 Placebo outcomes and alternative explanations

The paper uses several placebo and falsification strategies.

- **Private-grade components.** Exams and problem sets test whether unmarried women have lower ability or lower effort in general.
- **Writing ability.** This tests whether single women reduce all positive self-reports in public.
- **Comfort in competitive environments.** This helps show that not all job-related traits are affected; the response depends on what the trait signals in the marriage market.
- **Social-impact job choice.** In the supplementary experiment, the placebo job choice is unaffected by male peers.

Detailed interpretation: The results are strongest because the effects are not broad. They appear on outcomes that plausibly signal ambition or career intensity to potential partners.

4.6 Balance, inference, and robustness

- Randomization is checked for balance in observable characteristics.
- The paper reports robust standard errors in the primary experiment.
- In the supplementary experiment, inference is clustered at the group level.
- Because the number of groups is modest, the paper also reports permutation and wild-cluster-bootstrap style inference.
- Additional controls for observables do not remove the differential public effect for single women.

Detailed interpretation: These inference choices matter because the supplementary experiment has relatively few groups. The robustness checks protect against overstating precision.

4.7 What the design cannot fully prove

- The paper does not directly observe actual dating or marriage outcomes caused by the experimental responses.
- It does not observe students' sexual orientation, so the supplementary experiment is interpreted under the assumption that most single women are interested in male partners.
- It studies elite MBA students, so external validity to other populations is not guaranteed.
- It does not determine whether women have accurate beliefs about men's partner preferences.
- It cannot fully separate marriage-market preferences from broader gender-norm concerns, although the male-peer experiment strongly narrows the interpretation.

Detailed interpretation: These limitations do not invalidate the experiment. They define the scope: the paper identifies strategic public self-presentation in an elite MBA context, not the full long-run equilibrium of marriage and career outcomes.

4.8 Identification takeaway

The empirical strategy uses a layered design. The survey shows the mechanism is self-reported; grades show a visible/private behavioral contrast; the primary experiment shows that peer observability causally changes career self-presentation; and the supplementary experiment shows that male peer observability is the crucial audience.

5 Section-by-section study guide

5.1 Introduction

The introduction frames the paper around a trade-off between labor-market success and marriage-market desirability. It emphasizes that women may make career-relevant decisions while searching for partners and that actions such as speaking up, leading teams, negotiating, traveling, or working long hours can signal traits some men may not prefer in a partner.

Takeaway: Read the introduction as the conceptual bridge between labor economics and social-image economics. The paper is not mainly about discrimination by employers; it is about strategic behavior by women in anticipation of social and marriage-market judgments.

5.2 Section I: Observational evidence

Section I presents descriptive statistics, survey evidence, and grade evidence. The survey asks whether students avoided career-helping actions because of image concerns. The grade evidence compares visible participation with less-visible exams and problem sets.

Takeaway: Section I establishes plausibility. It does not by itself prove causality, but it shows that the behavior exists, that single women report the mechanism, and that visible behavior differs in the predicted group.

5.3 Section II: Experimental evidence

Section II is the heart of the paper. The primary experiment randomizes whether classmates are expected to observe career preferences. The supplementary experiment changes whether the audience contains male peers. These experiments are designed to move from suggestive observational patterns to causal evidence.

Takeaway: The primary experiment identifies the effect of public observability. The supplementary experiment identifies the relevant audience.

5.4 Interpretation section

The paper discusses alternative explanations: shyness, humility, ability differences, relationship-status selection, and generic public self-presentation. It argues that the pattern of treatment effects and placebos is hard to reconcile with these alternatives.

Takeaway: The interpretation section is important because the paper studies a sensitive mechanism. The authors do not rely on a single result; they use the full pattern across outcomes, groups, and audiences.

5.5 Conclusion

The conclusion emphasizes broader implications for gender gaps in schooling and careers. Many high-stakes decisions are made when people are single, and public visibility of ambition can matter in educational and workplace institutions.

Takeaway: The conclusion is not just about MBA students. The paper suggests a general mechanism through which social norms and partner-market incentives can shape economic investment.

6 Tables and figures

6.1 Table 1—Descriptive Statistics

Read:

- The grades sample contains 2,235 students across the 2010–2016 entering cohorts.
- The primary experiment sample contains 355 students from the 2016 cohort.
- The survey sample contains 272 students.
- Men make up nearly 70 percent of the grades sample, which matters because the marriage-market audience in the MBA program is male-heavy.
- Students are highly selected: the average GMAT score is high, and students have substantial pre-MBA work experience.
- Married and unmarried women differ somewhat in age and work experience, but single and non-single women in the experimental sample look more similar.
- The table defines the sample and clarifies why the paper uses married/unmarried in grades but single/non-single in the survey and experiments.

Detailed interpretation: Table 1 is the base for thinking about external validity and internal comparability. The sample is not a random cross-section of women; it is a high-ability, career-oriented MBA population. This makes the results especially striking because the women in the sample presumably care strongly about careers. At the same time, the selected sample means the paper should not be interpreted as measuring the average effect for all women in the economy.

The table also matters for identification. If single and non-single women looked very different on ability or career preparation, treatment heterogeneity could be less convincing. The table shows that the experimental comparison is more balanced than the married/unmarried administrative comparison.

Economic message: The experiment takes place in a market where both career incentives and marriage-market incentives are strong. That is why the MBA environment is a powerful setting.

6.2 Table 2—Avoidance of Workplace Behaviors by Gender and Relationship Status

Read:

- 73.1 percent of single women report avoiding at least one career-helping behavior to avoid appearing ambitious, assertive, or pushy.
- 63.5 percent of single women avoided taking initiative in negotiating a raise or promotion.
- 40.4 percent avoided asking for a leadership role.
- 25.0 percent avoided offering to make a presentation or sales pitch.
- 51.9 percent avoided speaking up at meetings.
- Non-single women report lower avoidance rates than single women on most behaviors.
- Single men and non-single men report much lower avoidance rates, especially for negotiation and speaking up.
- The p-values show that single women differ strongly from all other groups across most outcomes.

TABLE 1—DESCRIPTIVE STATISTICS							
	2010–2016 cohorts			2016 cohort			Survey
	Grades data			Primary experiment			
	Overall	Married women	Unmarried women	Overall	Non-single women	Single women	
Male (percent)	68.3			67.9			67.3
Age	28.0	28.9	27.1	27.9	27.5	27.3	28.1
Has children (percent)	4.7	15.9	0.2	5.7	0.0	1.7	
Years of work experience	5.1	5.8	4.6	5.2	5.0	4.9	
GMAT score	709	707	703	715	701	707	
<i>Marital/relationship status (self-reported, percent)</i>							
Single				46.2	0.0	100.0	53.3
In a serious relationship				22.0	51.9	0.0	18.0
Cohabiting				3.7	9.6	0.0	2.9
Engaged				5.9	15.4	0.0	4.8
Married				20.0	23.1	0.0	19.9
No response				2.3	0.0	0.0	1.1
<i>Marital/relationship status (admissions data, percent)</i>							
Single	77.0	0.0	100.0				
Married or in a domestic partnership	18.1	100.0	0.0				
No response	4.9	0.0	0.0				
<i>Citizenship (percent)</i>							
United States	66.0	51.2	68.2	62.5	71.2	58.3	
North America (without US)	4.1	7.3	2.4	3.9	5.8	3.3	
Asia	21.7	34.1	23.7	24.5	19.2	35.0	
Europe	3.8	2.4	2.9	2.0	1.9	1.7	
South America	3.7	4.9	1.9	6.2	1.9	1.7	
Africa	0.5	0.0	0.5	0.8	0.0	0.0	
Oceania	0.2	0.0	0.3	0.0	0.0	0.0	
Observations	2,235	82	582	355	52	60	272

Notes: The table shows descriptive statistics from the grades, primary experiment, and survey samples. The grades and primary experiment data are linked to admissions records. The survey was anonymous and cannot be linked to admissions records. Non-single refers to individuals who report being in a serious relationship, cohabiting, engaged, or married.

on the first day of the program at the career center introductory session, 98 percent of the class participated. The 2016 cohort looks similar to the earlier cohorts.

Admissions records for the 2010 to 2016 cohorts contain information on whether students were married or in a domestic partnership at the time they applied to the program: only 18 percent of students were. For the 2016 cohort, we collected more detailed information on students' relationship status during the primary experiment. While only 20 percent of students are married, less than one-half (46 percent) call themselves single. Twenty-two percent of students are in a serious relationship at the start of the MBA program, while just under 10 percent are cohabiting or engaged. These fractions are similar in the survey data (column 7). The survey was voluntary and conducted during the economics class during the middle of the semester and 76 percent of the class participated.¹¹ In the grades analysis, we can only compare

TABLE 2—AVOIDANCE OF WORKPLACE BEHAVIORS BY GENDER AND RELATIONSHIP STATUS (Survey Data)

	Taking initiative in negotiating a wage raise or promotion	Asking for a leadership role in a team or task force	Offering to make a presentation or sales pitch	Speaking up at meetings	Any behavior	Observations
Single women	63.5	40.4	25.0	51.9	73.1	52
Non-single women	39.4	24.2	15.2	33.3	60.6	33
Single men	25.3	23.0	18.4	27.6	43.7	87
Non-single men	30.3	23.6	6.7	29.2	50.6	89
<i>p-values of differences</i>						
Single versus non-single women	0.030	0.129	0.284	0.095	0.234	85
Single women versus others	0.000	0.014	0.031	0.002	0.002	261

Notes: Data are from a survey administered to first-year MBA students in the fall of 2016. Each number in the first four rows of data is the percentage of the group indicated by the row that avoided the action indicated by the column in their previous two years of work, despite the fact that they believed it could help their careers because they were concerned about appearing too ambitious, assertive, or pushy. Non-single refers to respondents in a serious relationship, cohabiting, engaged, or married.

single women were more likely to report having avoided these workplace behaviors, by amounts that are both economically and statistically significant. For example, 64 percent of single women did not take initiative in asking for a raise or a promotion because they were worried about looking too ambitious, relative to 39 percent of non-single women (the p -value of the difference is 0.030), 30 percent of non-single men, and 25 percent of single men (the p -value of the difference between single women and all other groups is below 0.001). Fifty-two percent

Detailed interpretation: Table 2 is a direct mechanism check. It asks students whether they themselves avoided career-helping actions for image reasons. The strongest result is negotiation avoidance, which is important because negotiating pay and promotions directly affects career advancement. The pattern also shows that the mechanism is not simply gender: non-single women are generally closer to men than to single women. The crucial interaction is gender plus relationship status.

The table also reveals that the behavior is not limited to one dramatic career decision. It covers everyday workplace actions: meetings, presentations, leadership roles, and bargaining. This supports the paper's claim that marriage-market incentives can distort many small labor-market investments.

Economic message: The survey shows that the mechanism is not hypothetical. Single women explicitly report avoiding valuable career actions because of how those actions would make them look.

6.3 Figure 1 and Table 3—Grades and Class Participation

Read:

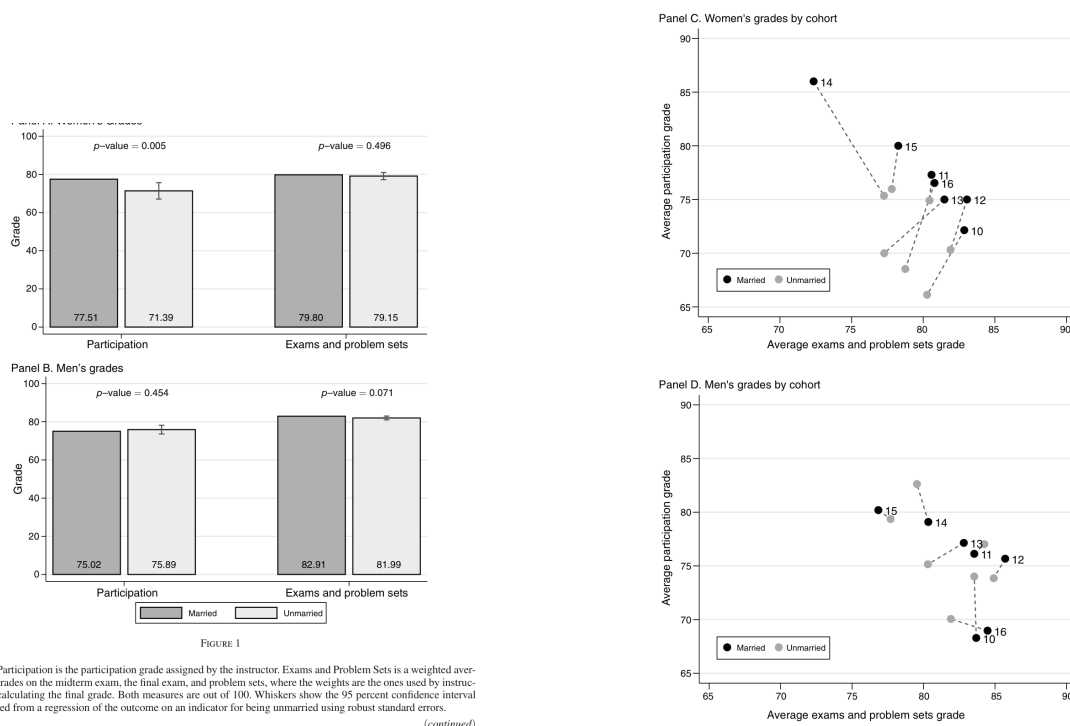
- Figure 1 Panel A compares married and unmarried women's participation and nonparticipation grades.
- Unmarried women score similarly to married women on nonparticipation grades: exams and problem sets.
- Unmarried women score much lower on class participation, a visible component of performance.
- Figure 1 Panel B shows that men do not display the same participation gap by marital status.
- Figure 1 Panels C and D plot residualized outcomes, showing that the participation gap for women remains after controlling for observables.
- Table 3 formalizes the relationship between relationship status and participation grades.
- The participation gap for unmarried women is about one-third of a standard deviation.

- The table shows that adding controls does not eliminate the gap.

Detailed interpretation: The grade evidence is powerful because it compares two types of performance within the same course. Exams and problem sets are academically important but relatively private. Participation is academically important and visible. The mechanism predicts that single or unmarried women should underperform only on the visible margin, not because they are less able, but because visible assertiveness has social or marriage-market costs.

Table 3 is not an experiment, so it cannot alone prove causality. But it makes the experimental evidence more credible by showing that the behavior appears in a real academic setting with actual grade consequences. The lack of a similar pattern for men is a crucial falsification test.

Economic message: Visibility changes incentives. A performance margin that requires public assertiveness can be distorted even when private ability is unchanged.



(insignificantly) positive. Panel C of Figure 1 shows that unmarried women have

(a) Figure 1, Panels A–B

(b) Figure 1, Panels C–D

Figure 1: Figure 1—Performance in Introductory Economics Class

6.4 Table 4—Effect of the Public Treatment on Reported Job Preferences and Skills

Read:

- Table 4 is the main causal table for the primary experiment.
- The public treatment lowers the career index for single women by about 0.56 standard deviations.
- The effect for single women is larger than the corresponding effects for non-single women, single men, and non-single men.

- The treatment lowers single women's desired compensation, willingness to travel, desired hours, tendency to lead, and professional ambition.
- The writing placebo is not affected, meaning the result is not simply public humility.
- The table includes controls and alternative specifications to show that the effect is not driven by observable imbalance.

Detailed interpretation: Table 4 is the cleanest evidence in the paper because treatment assignment is randomized. Students in both groups expect the career counselor to see their answers, so the professional incentive to look employable is constant. The only difference is whether classmates are expected to see the answers. Therefore, when single women reduce career-forward answers in public, the natural interpretation is strategic self-presentation to peers.

The index is important because the theory predicts a bundle of responses, not only one. A single coefficient on the index summarizes the overall move away from high-intensity career signals. The placebo outcome is equally important: writing ability is a positive skill, but it does not have the same marriage-market stigma. No treatment effect on writing makes the targeted signaling interpretation more persuasive.

Economic message: The treatment reveals a wedge between private career preferences and public career self-presentation. Gender gaps in public choices can emerge even when private choices look similar.

	Participation	Exams and problem sets	Midterm exam	Final exam	Problem sets
<i>Panel A. Women, no controls</i>					
Unmarried	−6.12 (2.19)	−0.65 (0.95)	0.26 (1.25)	−1.85 (1.63)	0.36 (0.28)
Dependent variable mean: Married women	77.51	79.80	82.41	68.22	95.70
Observations	664	664	664	664	664
R ²	0.01	0.00	0.00	0.00	0.00
<i>Panel B. Women, with controls</i>					
Unmarried	−5.81 (2.08)	−0.78 (0.94)	−1.28 (1.35)	−0.99 (1.55)	0.04 (0.23)
Dependent variable mean: Married women	77.13	80.10	82.86	68.57	95.79
Observations	644	644	644	644	644
R ²	0.28	0.25	0.17	0.31	0.53
<i>Panel C. Men, no controls</i>					
Unmarried	0.87 (1.16)	−0.92 (0.51)	−0.33 (0.74)	−1.89 (0.83)	0.05 (0.15)
Dependent variable mean: Married men	75.02	82.91	86.03	72.81	95.94
Observations	1,461	1,461	1,461	1,461	1,461
R ²	0.00	0.00	0.00	0.00	0.00
<i>Panel D. Men, with controls</i>					
Unmarried	−0.08 (1.15)	−1.64 (0.51)	−1.26 (0.76)	−2.94 (0.83)	0.05 (0.11)
Dependent variable mean: Married men	74.79	83.08	86.25	73.06	95.94
Observations	1,442	1,442	1,442	1,442	1,442
R ²	0.21	0.26	0.19	0.27	0.54

Notes: Each column in each panel shows the results of regressing the grade indicated by the column (out of 100) on a dummy for being not married or in a domestic partnership. Regressions in panels A and B are limited to women, while regressions in panels C and D are limited to men. Regressions in panels B and D include controls for age, GMAT score, years of work experience, US citizenship, and section fixed effects. Robust standard errors are in parentheses.

in participation grades by marital status is present throughout the GMAT score distribution and for both US and non-US citizens. It is stronger for younger women in the sample, but this may be in part because younger women are more likely to be truly single (not in long-term relationships), even among the unmarried set. Because some unmarried students are actually in relationships, the true difference in participation grades between single and non-single women may be larger than what we observe.

While professors have discretion over participation grades and these grades are by their nature subjective, it is unlikely that our results are driven by professors discriminating against unmarried women. Unconscious bias could lead professors to give women lower participation grades than men for the same comments. However, it seems less likely that professors would discriminate against unmarried women relative to married women. Married women have participation grades that are higher than their male counterparts. In personal correspondence, professors reported not knowing their students' relationship status. Additionally, the point estimates are very

	Kling-Liebowitz index	Desired compensation	Days per month of travel	Desired weekly hours of work	Tendency to lead	Professional ambition	Comfort in competitive environments	Writing skills
<i>Panel A. Single women</i>								
Public treatment	−0.56 (0.13)	−18.12 (8.17)	−6.93 (2.35)	−3.89 (2.11)	−0.39 (0.19)	−0.75 (0.18)	0.12 (0.21)	0.13 (0.23)
Private treatment mean	−0.06	131.05	13.55	52.21	3.87	4.13	3.29	3.84
Observations	59	60	60	59	60	60	60	60
R ²	0.23	0.08	0.13	0.05	0.07	0.23	0.01	0.01
<i>Panel B. Non-single women</i>								
Public treatment	−0.15 (0.14)	−1.22 (7.77)	0.65 (3.14)	−4.06 (1.87)	−0.05 (0.18)	−0.14 (0.19)	−0.09 (0.24)	−0.07 (0.17)
Private treatment mean	0.00	134.72	9.67	52.54	3.89	4.26	3.63	4.11
Observations	51	52	52	52	52	52	51	52
R ²	0.02	0.00	0.00	0.08	0.00	0.01	0.00	0.00
<i>Panel C. Single men</i>								
Public treatment	0.04 (0.12)	−0.89 (7.57)	2.72 (2.36)	0.39 (2.09)	0.15 (0.15)	−0.07 (0.17)	−0.11 (0.15)	0.03 (0.18)
Private treatment mean	0.15	146.88	15.38	52.25	3.69	4.23	4.02	3.90
Observations	103	104	103	104	104	104	104	104
R ²	0.00	0.00	0.01	0.00	0.01	0.00	0.01	0.00
<i>Panel D. Non-single men</i>								
Public treatment	0.09 (0.10)	−7.13 (6.08)	2.39 (1.94)	3.34 (1.78)	0.11 (0.13)	−0.02 (0.12)	0.00 (0.14)	−0.12 (0.17)
Private treatment mean	−0.05	140.86	9.94	51.14	3.75	4.15	3.67	3.82
Observations	130	130	131	131	131	131	131	131
R ²	0.01	0.01	0.01	0.03	0.01	0.00	0.00	0.00
<i>Panel E. p-values: difference in the effect of the public treatment</i>								
Single versus non-single women	0.032	0.129	0.050	0.952	0.191	0.018	0.494	0.490
Single women versus others	0.000	0.101	0.000	0.042	0.018	0.000	0.471	0.475

Notes: Each cell in panels A through D presents the results of regressing the outcome indicated by the column on a dummy for being in the public treatment. Regressions are limited to the sample indicated by the panel. No controls are included. The Kling-Liebowitz index is defined in the text. The desired compensation and hours of work variables correspond to the midpoint of the range the respondent chose. Desired compensation is in thousands of dollars. The travel variable is the number of days per month the respondent would be willing to travel; it is also coded as the midpoint of the chosen range. The remaining outcomes are on a 1-to-5 scale. Robust standard errors are in parentheses. Panel E provides for each outcome the p-values for the tests that the effect of the public treatment is the same for (i) single and non-single women and (ii) single women and all other students.

and non-single women respond differently when they believe their answers will be public, but they behave similarly in private. This suggests that single and non-single women have similar career goals and would like to send similar signals to their career advisor, but that single women face a more significant trade-off between signaling their career advisor and their classmates. Online Appendix Table 4 shows these results do not change when we include controls for student

6.5 Figures 2–4—Desired Compensation, Travel, and Weekly Hours

Read:

- Figure 2 shows desired compensation. Single women report substantially lower desired compensation in the public treatment than in the private treatment.

- The desired-compensation gap for single women is economically meaningful because salary expectations affect matching to job tracks.
- Figure 3 shows willingness to travel. Single women sharply reduce reported willingness to travel when public.
- Figure 4 shows desired weekly hours. Single women report fewer desired weekly hours in public.
- Non-single women and men generally do not show the same pattern across these outcomes.
- These outcomes are important for MBA recruiting because they influence whether students are steered toward consulting, finance, investment banking, or less travel-intensive jobs.

Detailed interpretation: These figures move the result from an abstract index to concrete job-market decisions. Compensation, travel, and hours are not symbolic traits; they are inputs into internship and job placement. Lower desired compensation can make a candidate look less aggressive or less career-oriented. Lower willingness to travel or work long hours can shift the set of jobs viewed as suitable.

The private-treatment comparisons are also important. Single women do not look systematically less career-oriented in private. The gender gap emerges when choices are public. This supports the idea that public observability creates the gap rather than merely revealing preexisting preferences.

Economic message: The public treatment can change the labor-market opportunities students are matched to. Marriage-market signaling can therefore have real career consequences.

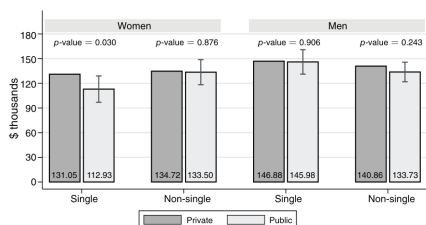


FIGURE 2. DESIRED COMPENSATION (Primary Experiment)

Notes: Students were asked their desired compensation in their first year after graduation, including base pay, performance pay, and equity, but excluding signing bonus. Desired compensation is coded as the midpoint of the chosen range, except for "under \$75,000" (coded as \$62,500) and "above \$250,000" (coded as \$262,500). Some

(a) Figure 2—Desired Compensation

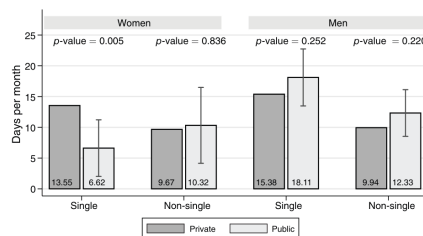


FIGURE 3. DAYS PER MONTH WILLING TO TRAVEL (Primary Experiment)

Notes: Students were asked how often they are willing to travel for work. Willingness to travel is coded as the midpoint of the chosen range, except for "rather not travel" (coded as 0) and "as much as necessary" (coded as 30). Whiskers show the 95 percent confidence interval calculated from regressions of the number of days per month the

(b) Figure 3—Days per Month Willing to Travel

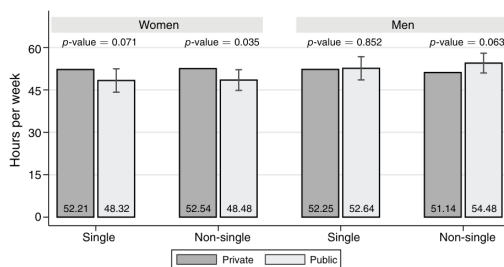


FIGURE 4. DESIRED WEEKLY HOURS OF WORK (Primary Experiment)

Notes: Students were asked how many hours per week they are willing to work on a regular basis. Desired hours of work is coded as the midpoint of the chosen range, except for "over 80" (coded as 85.5, which would be the midpoint of an 81 to 90 hour range, since ranges are 41–50 hours, 51–60 hours, etc.). Some respondents chose two or more consecutive answers. Their responses are coded as the midpoint of the full range chosen. Whiskers show the

Figure 3: Figure 4—Desired Weekly Hours of Work

6.6 Figures 5–8—Leadership, Ambition, Competition, and Writing Placebo

Read:

- Figure 5 shows tendency to lead. Single women rate themselves lower in public than in private.
- Figure 6 shows professional ambition. The public treatment strongly reduces single women’s reported ambition.
- Figure 7 shows comfort in competitive environments. The public/private difference is not large for single women.
- Figure 8 shows writing ability. This placebo outcome is stable across private and public treatments.
- The largest self-rating effect is professional ambition, which is precisely the trait the theory predicts may be costly in the marriage market.
- The absence of an effect on writing ability shows the public treatment does not make single women generally understate all positive traits.

Detailed interpretation: The self-rating figures show the mechanism at the level of traits. Leadership and ambition are traits employers may value but some partners may view as threatening or undesirable. Writing ability is a skill employers value, but it does not carry the same marriage-market penalty. The contrast between ambition/leadership and writing is therefore central.

The comfort-in-competitive-environments result is useful because it shows that the paper’s predictions depend on perceived social meaning. If respondents or male observers do not interpret that question as a negative partner signal, we should not expect a strong treatment effect. This nuance strengthens rather than weakens the interpretation.

Economic message: The distortion is not general modesty. It is targeted management of specific signals that combine labor-market value with marriage-market cost.

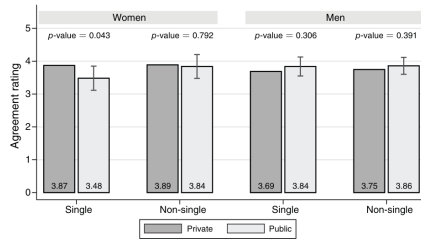


FIGURE 5. TENDENCY TO LEAD (Primary Experiment)

Notes: Students rated agreement with the statement “You tend to lead in your day-to-day interactions” on a 1–5 scale, where 1 is Strongly Disagree and 5 is Strongly Agree. Whiskers show the 95 percent confidence interval

(a) Figure 5—Tendency to Lead

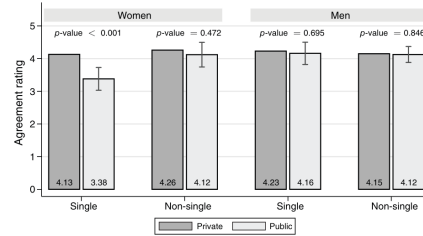


FIGURE 6. PROFESSIONAL AMBITION (Primary Experiment)

Notes: Students rated agreement with the statement “You are more professionally ambitious than your most recent work colleagues” on a 1-to-5 scale, where 1 is Strongly Disagree and 5 is Strongly Agree. Whiskers show the 95 percent confidence interval calculated from regressions of students’ answers on an indicator for being in the

(b) Figure 6—Professional Ambition

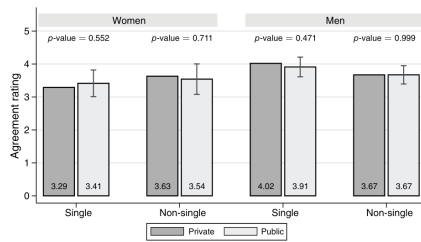


FIGURE 7. COMFORT IN COMPETITIVE ENVIRONMENTS (Primary Experiment)

Notes: Students rated agreement with the statement “You feel very comfortable in competitive environments” on a 1-to-5 scale, where 1 is Strongly Disagree and 5 is Strongly Agree. Whiskers show the 95 percent confidence interval.

(a) Figure 7—Comfort in Competitive Environments

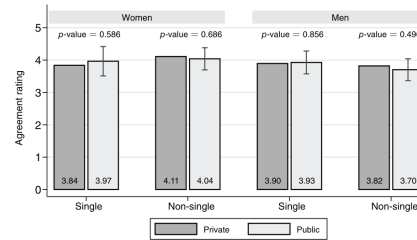


FIGURE 8. WRITING ABILITY (Primary Experiment)

Notes: Students rated agreement with the statement “You have above-average writing skills” on a 1-to-5 scale, where 1 is Strongly Disagree and 5 is Strongly Agree. Whiskers show the 95 percent confidence interval calculated from regressions of students’ answers on an indicator for being in the public treatment using robust standard errors. Non-single respondents are in a serious relationship, cohabiting, engaged, or married.

(b) Figure 8—Writing Ability

6.7 Figure 9 and Table 5—Group Gender Composition and Reported Job Preferences

Read:

- Figure 9 focuses only on single women in the supplementary experiment.
- In all-female groups, 68 percent choose the higher-salary job over the shorter-hours job.
- In groups with male peers, this falls to 43 percent.
- In all-female groups, 79 percent choose better promotion opportunities over less travel.
- In groups with male peers, this falls to 37 percent.
- The placebo choice, social impact over coworker interaction, is essentially unchanged.
- Table 5 estimates these differences with section fixed effects and also studies heterogeneity by the share of unmarried men in the group.

Detailed interpretation: This is the strongest direct evidence that the relevant audience is the marriage market. The treatment does not merely make choices public; it changes who observes them. The career-intensive choices decline when male peers are present, while the placebo choice does not. This directly supports the idea that single women are managing how they look to potential male partners.

The use of section fixed effects matters because randomization occurs within sections. The paper also uses permutation and bootstrap methods because the number of groups is limited. This strengthens confidence that the result is not driven by conventional standard-error problems.

Economic message: The same woman can express different career preferences depending on whether the audience is potential romantic partners. The audience, not just visibility, matters.

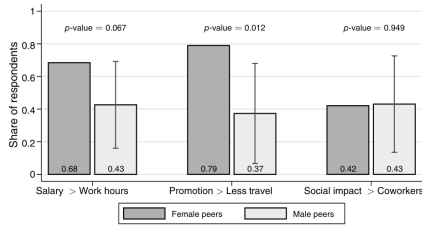


FIGURE 9. GROUP GENDER COMPOSITION AND REPORTED JOB PREFERENCES
(Supplementary Experiment, Single Women Only)

Notes: Each set of bars represents the fraction of single women who reported a preference for (i) a job with a higher salary over a job with shorter work hours, (ii) a job with better promotion opportunities over a job with less travel, or (iii) a job with a social impact over a job with more interactions with coworkers. The "Female Peers" bars show

(a) Figure 9—Group Gender Composition and Reported Job Preferences

TABLE 5—EFFECT OF GROUP COMPOSITION ON SINGLE WOMEN'S REPORTED JOB PREFERENCES
(Supplementary Experiment)

	Kling-Liebm-Katz index	Prefers higher salary over fewer hours	Prefers promotion over less travel	Prefers social impact over interactions with coworkers
<i>Panel A. Peer gender</i>				
Male peers indicator	−0.77 (0.23)	−0.26 (0.14)	−0.42 (0.16)	0.01 (0.15)
Mean for single women in female groups	0.00	0.68	0.79	0.42
Observations	40	40	40	40
R ²	0.29	0.14	0.26	0.09
<i>Panel B. Marital status of male peers</i>				
Share of male peers who are unmarried	−1.20 (0.34)	−1.23 (0.19)	0.08 (0.27)	0.44 (0.33)
Mean for single women in male groups	−0.58	0.43	0.52	0.38
Observations	21	21	21	21
R ²	0.36	0.61	0.19	0.22

Notes: Panel A shows the results of regressing either the Kling-Liebm-Katz index or an indicator for choosing a given job on a dummy for being in a group with male peers, controlling for section fixed effects. Panel B shows the results of regressing the same dependent variables on the share of male peers who are unmarried. Regressions are limited to single women in panel A and to single women in groups with male peers in panel B. The choices presented and the Kling-Liebm-Katz index are described in the text. Standard errors clustered at the group level are in parentheses.

The first question asked students to choose between a job with a high salary requiring 55–60 hours per week and a job with a lower salary requiring 45–50 hours

(b) Table 5—Effect of Group Composition on Single Women's Reported Job Preferences

7 Short summary

- The paper studies whether marriage-market incentives affect women's labor-market investments.
- The setting is an elite MBA program where students are making career investments and interacting with potential partners.
- Single women report avoiding workplace behaviors that could help their careers to avoid looking ambitious, assertive, or pushy.
- The survey evidence is strongest for negotiation and speaking up, two behaviors with direct career consequences.
- In administrative grades, unmarried women perform similarly to married women on private/nonvisible grade components.
- Unmarried women perform substantially worse on class participation, which is visible to peers.
- Men do not show the same relationship-status pattern in grades.
- The primary experiment randomizes whether classmates are expected to observe career questionnaire responses.
- Single women lower desired compensation, willingness to travel, desired weekly hours, tendency to lead, and professional ambition in the public treatment.
- Men and non-single women do not respond in the same way.
- Writing ability is unaffected, ruling out a simple public-humility explanation.
- The supplementary experiment shows that single women are less career-focused when observed by male peers rather than only female peers.
- The results suggest that public visibility of ambition can create gendered distortions in career investment.
- The effects occur in a highly selected group that likely values career success, suggesting the mechanism may be relevant beyond this setting.
- The paper contributes to labor economics, gender economics, and social-image economics by showing that partner-market incentives can shape labor-market behavior.

8 Conclusion

8.1 Core conclusion

Bursztyn, Fujiwara, and Pallais show that marriage-market incentives can distort labor-market investments. Single women in an elite MBA program strategically downplay ambition and career intensity when those traits are visible to classmates, especially male classmates. This behavior is not observed for men or non-single women, and it does not appear on placebo traits such as writing ability.

8.2 Economic intuition

The core economic intuition is that individuals often operate in multiple markets at once. An action that improves outcomes in one market can damage outcomes in another. For single women in this setting, ambition, leadership, willingness to travel, and willingness to work long hours can help in the labor market but may be costly in the dating or marriage market. If the action is private, the labor-market payoff dominates. If the action is public to potential partners, the marriage-market cost can change behavior.

8.3 Why the evidence is persuasive

The evidence is persuasive because the paper uses several complementary strategies. The survey establishes that single women report the mechanism directly. Grade data show a real behavioral gap on a visible performance margin but not on private performance margins. The primary experiment causally changes observability while holding career-counselor observability fixed. The supplementary experiment changes the gender composition of the observing peer group. Finally, placebo outcomes show that the results are not simply about humility or generalized self-presentation.

8.4 Implications for labor-market inequality

The results imply that gender gaps can emerge before formal labor-market discrimination or family constraints. If women avoid visible ambition while single, they may enter the labor market with lower leadership visibility, lower willingness to pursue high-hours/high-travel careers, weaker negotiation history, and fewer high-intensity career signals. These choices can compound over time because early internships, first jobs, and early promotions have long-run consequences.

8.5 Implications for education and workplace design

The paper suggests that institutions should think carefully about which actions are public. Participation grades, public leadership contests, public preference forms, public negotiations, and visible self-promotion can create asymmetric social costs. Anonymized preference collection, structured participation systems, cold-calling rather than voluntary participation, and private career counseling may reduce distortions.

8.6 Relation to gender norms and identity

The results are consistent with gender norms that penalize women for traits rewarded in men. The evidence does not require that women personally prefer lower ambition; in private, single women often look similar to other groups. The distortion arises when public signals interact with social expectations about what a desirable wife or partner should be like.

8.7 Limitations and open questions

The sample is an elite MBA cohort, so external validity is not automatic. The paper does not directly observe long-run job placement, wages, promotions, dating outcomes, or marriages caused by the experimental treatments. It also does not observe sexual orientation. Another open question is whether women accurately perceive men's preferences or whether they overestimate the marriage-market penalty for ambition. Future work could measure long-run effects, study other populations, and test interventions that reduce the social cost of ambition.

8.8 Final takeaway

The paper's central lesson is that gender gaps can be created by visibility itself. When ambition is observable to potential partners, some women strategically hide it. Policies and institutions that ignore this visibility channel may unintentionally reward those who can display ambition without social penalty and penalize those who cannot.